A Reliability-Aware Comparison of Aggregation Methods for Catalog-Recorded Descriptor Prediction in Two-Molecule Fragrance Blends

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Bachelor of Science degrees

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We declare that the Senior Project entitled

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which we submitted to the faculty of the

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is our own work. To the best of our knowledge, it does not contain materials published or written by another person, except where due citation and acknowledgement is made in our senior project documentation. The contributions of other people whom we worked with to complete this senior project are explicitly cited and acknowledged in our senior project documentation.

We also declare that the intellectual content of this senior project is the product of our own work. We conceptualized, designed, encoded, and debugged the source code of the core programs in our senior project. The source code of third party APIs and library functions used in our program are explicitly cited and acknowledged in our senior project documentation. Also duly acknowledged are the assistance of others in minor details of editing and reproduction of the documentation.

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ABSTRACT

Combining two molecular representations is an unordered-pair learning problem: exchanging the two inputs should not change the prediction, while the model may still need to preserve relationships between them. This study will compare mean pooling, order-averaged concatenation, and Set Transformer aggregation for multi-label prediction of catalog-recorded descriptors of two-molecule blends. The three variants will use a shared molecular graph encoder and matched training conditions. Order-averaged concatenation will be treated as an adapted baseline based on a published pairwise concatenation design, not as a full reproduction of that model.

The catalog descriptors will be treated as proxy targets rather than controlled sensory-panel measurements. Reliability-aware eligibility masking and component-disjoint partitioning will be used as evaluation controls: the first will account for uncertain catalog annotations, while the second will reduce repeated molecular components across data partitions. The models will be compared on reliability-screened test cells using macro F1 as the primary metric and micro F1, PR-AUC, and Hamming loss as supporting metrics. Additional analyses will examine catalog-label patterns and how the relative performance of the three methods changes when training-label reliability is reduced, including under controlled label corruption. Numerical results will be reported only after the finalized evaluation pipeline has been run.

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Chapter 1

Introduction

Combining two graph-derived vectors is an unordered-pair learning problem. Each molecule can be encoded as a graph and represented by a learned vector, but a blend should produce the same prediction whether the inputs are presented as (A, B) or (B, A). The combination method must therefore preserve input-order invariance while retaining information about the particular pair being modeled. Mean pooling is a simple order-independent baseline, but it can map different pairs to the same representation: the pairs $(2, 8)$ and $(5, 5)$ have the same average even though their elements differ. More expressive aggregation may preserve distinctions that averaging loses, but it also adds parameters and computation. Whether that added complexity produces a useful improvement is an empirical question, not an assumption of this study [41].

Fragrance-blend prediction provides the application setting for this pairwise learning problem. Graph Neural Networks (GNNs) have been used to learn molecular representations directly from structure, including in single-molecule odor prediction [23, 27]. Extending a single-molecule encoder to a two-molecule input requires a separate aggregation design. This study compares mean pooling, order-averaged concatenation, and Set Transformer aggregation for predicting catalog-recorded descriptors of binary fragrance blends [24, 37].

The blend descriptors used for training and evaluation are catalog records, not direct measurements of human perception. Public blend datasets such as the one compiled by Sisson et al. [37] are based on reference-catalog recommendations rather than a controlled sensory panel. The study therefore measures label consistency using reciprocal and cross-catalog agreement before comparing

the aggregation methods, and reports performance under declared reliability conditions rather than treating every catalog label as unquestioned ground truth.

1.1 Project Context

The fragrance domain supplies a useful application for studying this representation problem. Existing molecular GNNs can encode individual structures, but a pairwise blend task requires a second design decision: how to combine two molecule embeddings into one order-independent representation before multi-label classification [23, 37]. The unresolved issue addressed here is therefore not computational olfaction in general. It is the controlled comparison of alternative pair-combination methods under the same encoder, training conditions, and evaluation protocol.

The data structure also creates a partitioning problem. A blend dataset is organized as pair records, and the same molecule can occur in many different pairs. A random split of pair records can therefore place one molecule in both training and test partitions, allowing the model to reuse a component it has already seen. This study uses a component-disjoint split and graph-carving procedure to reduce that overlap by excluding pairs that cross the assigned partitions. The procedure is intended to reduce component leakage; it does not guarantee that every held-out molecule belongs to an entirely unfamiliar chemical family [12].

The labels create a second evaluation problem. Each blend may have several descriptors, but only a small fraction of the possible descriptor cells are positive. A model can therefore obtain a deceptively favorable aggregate score by predicting mostly absent labels, while inconsistent catalog records can change the apparent ranking of the aggregation methods. The study addresses these risks through multi-label metrics suited to sparse outputs and through a reliability audit based on reciprocal listing and cross-catalog agreement. The audit is used to define declared reliability conditions; it does not convert catalog records into verified measurements of human perception.

Fragrance and catalog terminology are retained only to identify the input domain and target descriptors. The study does not attempt to infer receptor mechanisms, explain physical mixture chemistry, or model perfume-formulation practice. Its computer-science focus is the representation and evaluation of an unordered pair of molecular graphs.

1.2 Purpose and Description

This study contributes a controlled empirical comparison of mean pooling, order-averaged concatenation, and Set Transformer aggregation under a shared molecular-graph encoding framework. It does not propose a new neural architecture. Instead, it evaluates whether an existing attention-based set aggregator provides a measurable benefit over simpler alternatives for predicting catalog-recorded descriptors of two-molecule blends [24, 37].

In all three variants, a Graph Attention Network (GAT) encodes each molecular graph into a learned vector using the same architectural specification and initialization policy [49]. The experimental variants differ in how the two molecule vectors are combined. Mean pooling averages the vectors directly. Order-averaged concatenation evaluates both input orders and averages their outputs. Set Transformer aggregation uses Self-Attention Blocks (SAB) and Pooling by Multihead Attention (PMA) to learn statistical relationships between the two vectors before producing an order-independent blend representation [24, 51].

A shared multi-label classification design maps each combined representation to independent sigmoid outputs for the target catalog descriptors. The variants are compared using matched data partitions, training conditions, and evaluation metrics. Reliability-aware evaluation reduces the influence of uncertain catalog annotations, while component-disjoint evaluation reduces the risk that the same molecular component appears in both training and test pairs. These controls strengthen the comparison, but they do not establish label correctness, eliminate all structural similarity, or guarantee that the Set Transformer will outperform the simpler methods.

Significance of the Study

The study is primarily intended to contribute evidence for computer-science research on pairwise representation learning and model evaluation. Its findings may be useful to the following groups:

- **Machine-Learning and Graph-Representation Researchers.** The study provides a controlled comparison of three methods for combining an unordered pair of graph embeddings. It contributes evidence on whether an attention-based set aggregator offers a measurable benefit over simpler methods, how model comparisons change under uncertain labels, and how component-disjoint splitting can reduce the influence of repeated molecular components across data partitions. Computational olfaction serves as the test domain for this broader represen-

tation and evaluation problem.

- **Researchers and Maintainers of Olfactory Reference Catalogs.** The reciprocal and cross-catalog agreement analyses provide a way to describe the consistency of catalog-derived labels before they are used for machine learning. These analyses may help identify descriptors that require closer review, although they do not determine whether a catalog label is perceptually correct.

- **Fragrance Researchers and Practitioners.** The results may provide preliminary evidence about the strengths and limits of structure-based prediction for catalog-recorded blend descriptors. However, this study does not validate a formulation tool or measure reductions in laboratory work, material use, or trial-and-error blending [31].

Research Questions

This study seeks to answer the following questions:

RQ1. How do mean pooling, order-averaged concatenation, and Set Transformer aggregation compare in predicting reliable catalog-recorded descriptors of two-molecule blends under the same training and evaluation conditions?

RQ2. How does the performance of the three aggregation methods differ across additive, catalog-label loss, and catalog-label gain groups?

RQ3. How does changing the reliability of the training labels, including through controlled label corruption, affect the performance of the three aggregation methods?

The study also has a limited connection to the United Nations Sustainable Development Goals. Its work on computational methods and data-quality evaluation is most closely related to SDG 9. Structure-based prediction could eventually support tools related to responsible material use under SDG 12, but the present study does not measure material savings, reduced laboratory work, or reduced trial-and-error formulation. These are possible future applications, not outcomes evaluated in this research.

1.3 Objectives

General Objective

To compare mean pooling, order-averaged concatenation, and Set Transformer aggregation as methods for combining an unordered pair of molecular graph embeddings for multi-label prediction of catalog-recorded blend descriptors, using matched training conditions and a reliability-aware, component-disjoint evaluation protocol.

To achieve the general objective, the following specific objectives must be accomplished:

- To prepare the single-molecule and blend datasets and describe the consistency and sparsity of their catalog-recorded labels.
- To construct fixed component-disjoint data partitions and reliability-screened evaluation conditions for comparing the models.
- To implement mean pooling, order-averaged concatenation, and Set Transformer aggregation using the same GAT encoder design, output task, loss function, and matched training protocol.
- To compare the three aggregation methods using macro F1 as the primary metric and micro F1, PR-AUC, and Hamming loss as supporting metrics.
- To determine how the relative performance of the three methods changes across label-reliability conditions, controlled label corruption, and additive, catalog-label loss, and catalog-label gain patterns.

1.4 Scope and Limitations

This study is limited to pairwise, two-molecule blends and the prediction of catalog-recorded descriptors. Each molecule is represented as a two-dimensional molecular graph and encoded by a shared Graph Attention Network (GAT) [49]. The study compares mean pooling, order-averaged concatenation, and Set Transformer aggregation as three methods for combining the resulting molecule embeddings. The Set Transformer uses attention to learn statistical relationships between the embeddings; it is not treated as a model of receptor activity or physical molecular interaction [24, 51].

The data are obtained from the GoodScents and Leffingwell single-molecule catalogs and the catalog-derived binary-blend dataset of Sisson et al. [7, 37]. The target descriptors are binary catalog records rather than intensity ratings or measurements from a newly conducted sensory panel. The study does not recruit human assessors and does not claim that its predictions represent what every person would perceive. Concentration, mixing ratio, and changes over time are also outside the scope; each blend is treated as an unweighted pair.

Multi-label learning requires both positive and negative targets. A descriptor recorded for a blend is treated as positive, while an unlisted descriptor is treated as negative under a closed-world assumption. This is an operational definition for model training and evaluation. Catalog omission may reflect an unrecorded observation rather than true sensory absence, so an unlisted cell is not treated as proof that the descriptor cannot be perceived.

Label reliability is assessed using reciprocal listing and cross-catalog agreement. These checks measure the consistency of the available records, not their perceptual truth. Reliability-screened training and evaluation use a cell-level eligibility mask: reliable recorded positives and eligible unlisted negatives are included, while ambiguous cells are excluded. This procedure can reduce the influence of uncertain annotations, but it cannot remove every form of catalog bias or labeling error.

The models are evaluated on fixed component-disjoint partitions produced by the clustering and graph-carving procedure described in Chapter 4. Pairs that cross partition boundaries are excluded so that a retained training pair and a retained test pair do not share the same molecule. This design reduces component leakage, but it does not guarantee that the partitions have no structural similarity or that the test molecules represent every possible unseen chemical family [12].

The comparison is limited to the declared model variants, metrics, reliability conditions, and controlled label-corruption experiments. The results therefore show how these methods behave on the selected catalog-derived task and should not be generalized directly to larger mixtures, other descriptor vocabularies, or human sensory judgments. Order-averaged concatenation is an adapted baseline based on the pairwise design of Sisson et al.; it is not a full reproduction of that study's model. Final dataset sizes, target-vocabulary size, and label density will be reported only after the revised data pipeline has been run.

1.5 Definition of Terms

Molecular Graph

A computational representation of a molecule in which atoms are nodes, bonds are edges, and both carry features used by the graph encoder.

Graph Embedding

A fixed-length numerical vector produced by the graph encoder to summarize the learned information from one molecular graph.

Multi-Label Classification

A prediction task in which one blend can be assigned several descriptors at the same time. Each target descriptor is predicted separately as present or absent.

Mean Pooling

A combination method that takes the element-wise average of the two molecule embeddings. It is naturally independent of input order, but different pairs can produce the same average.

Order-Averaged Concatenation

A combination method that places the two molecule embeddings side by side, evaluates both input orders, and averages the two model outputs so that the final prediction does not depend on which molecule is listed first.

Set Transformer Aggregation

An attention-based method for combining an unordered collection of embeddings. In this study, it learns statistical relationships between the two molecule embeddings before pooling them into one blend representation.

Permutation or Order Invariance

The requirement that exchanging the positions of the two input molecules must not change the model's final prediction.

Component-Disjoint Split

A data partition in which the same molecule does not occur in both a retained training pair and a retained test pair. It is designed to reduce component leakage, but it does not guarantee that the

partitions contain no structurally similar molecules.

Reliability Mask

A cell-level indicator that determines which blend–descriptor targets are included in reliability-screened training or evaluation. Reliable recorded positives and eligible unlisted negatives are included, while ambiguous cells are excluded.

Evaluation Metrics

Macro F1 gives each retained descriptor equal weight when averaging its F1 score. Micro F1 combines all evaluated decisions before computing one F1 score. Precision–Recall Area Under the Curve (PR-AUC) measures how well the model ranks positive targets across decision thresholds. Hamming loss reports the fraction of evaluated present-or-absent decisions that are incorrect.

Chapter 2

Review of Related Literature

This chapter reviews papers that help build the idea for this study. It begins with graph-based scent prediction for single molecules, then summarizes what is known about how odor mixtures behave to be able to explain the inputs and the labels this study has. Next, it discusses how an unordered pair of learned vectors can be combined, which covers the three methods this paper will be comparing: mean pooling, order-averaged concatenation, and Set Transformer aggregation, together with the precedents in mixture modeling. The chapter then reviews the catalog-derived datasets and talks about the problems regarding this study's labels, particularly their reliability, the noise, and their sparsity. This is followed by the data-splitting and model-comparison methods that help make evaluation credible. It ends by synthesizing the literature into the controlled-evaluation gap that this thesis addresses.

2.1 Computational Olfaction and Single-Molecule Prediction

This first section explores the evolution of computational olfaction from early QSOR modeling with manual feature engineering to more modern graph-based methods. It reviews how these frameworks were used in predicting scent from a single molecule and the limitations of such when applied to the multi-molecule blend prediction task this thesis aims to achieve.

2.1.1 History of QSOR Modeling

Quantitative Structure-Odor Relationship (QSOR) modeling refers to the prediction of the odors and olfactory properties of molecules through the characterization of their structure [14, 47, 52]. QSOR is especially important in fields such as the fragrance and flavor industries, as it could help reduce the trial-and-error involved in the formulation and development of new compounds computationally, as experimental wet lab methods can be both time-consuming and expensive [19, 33].

Early approaches in QSOR involved the manual creation of numerical features that describe and quantify the properties of a molecule [3, 14, 53]. These included molecular fingerprints, which encode the presence or absence of substructures in a binary vector, and physicochemical descriptors computed with dedicated software [3, 10, 35, 46].

Many studies pair such descriptors or fingerprints with classical learners such as random forest, XGBoost, and LightGBM to predict odor [33, 40, 45]. Two comparisons are representative. Suh et al. found that Morgan-fingerprint models outperformed descriptor-based ones on a curated set of 8,681 fragrance compounds [40], and Achebouche et al. found that graph convolutional representations outperformed general molecular descriptors and fingerprints on a set of 5,955 molecules [1].

While effective, these methods require the features to be chosen in advance [4, 23, 53], so the learning model works from a fixed summary of each molecule. Graph neural networks address this by learning, from the atoms and bonds themselves, which combinations of features are most useful for the task, although people still choose the input features and the network design [2, 4, 17, 52]. The best-known graph-based odor model is discussed next.

2.1.2 The Principal Odor Map

Addressing the aforementioned limitation is Lee et al.'s Principal Odor Map (POM), which maps a molecule's chemical structure to its smell using Graph Neural Network (GNN) techniques [23]. Rather than evaluating a molecule's scent using a static representation built from predefined hand-crafted descriptors, POM learns directly from the molecule's structure during training. It does this by treating each molecule as a graph, where nodes and edges represent atoms and chemical bonds, respectively; each node and edge is itself a vector, with nodes storing properties such as atomic number, and edges storing bond properties. Through message passing, the GNN dynamically assigns weights to each of the molecule's structural features, producing a numerical representation of the

molecule [23].

The setup was as follows: Lee et al. used approximately 5,000 molecules derived from merging data from the GoodScents and Leffingwell fragrance databases to produce a multi-label output of 138 distinct semantic odor descriptors. It was then tested on new molecules against a human benchmark, wherein approximately 15 trained panelists rated molecules using a narrowed-down set of 55 odor descriptors. The model's predictions fell within the range of individual panelists' ratings for every label except musk, and surpassed the median panelist on 30 out of the 55 labels (55%) [23]. This study differs from theirs because they checked their model against a newly collected human panel, while this study has no panel and evaluates against catalog-recorded labels only.

This shows that a graph model trained on catalog labels can predict panel-rated scent from molecular structure alone at roughly the level of a typical panelist. However, despite this significant advancement, these results were derived exclusively from single, unmixed molecules.

2.1.3 Limitations of Single-Molecule Models

POM shows what a graph encoder can learn for one molecule, but it operates on a single molecule at a time, while odorants rarely appear in isolation [8, 23, 43, 50]. In the real world, scents are rarely composed of single molecules and are instead blends that can contain individual components numbering up to the hundreds [16, 50]. A single-graph encoder produces one vector per molecule but it does not define how the vectors of two molecules should be combined into one prediction. That combining step is the subject of this study.

Lee et al. admit this limitation themselves, noting that while POM uses a coordinate system to categorize scents, it alone cannot determine the behavior and outcomes of different molecules when mixed [23].

2.2 Psychophysics of Odor Mixtures

Because single-molecule models do not specify how two molecules are combined, a method for combining them is needed. This section establishes only that both simple averaging and learned aggregation are plausible candidates for that method. It reviews how catalog blend labels differ from the labels of their parent molecules, and the human-panel evidence that averaging may be sufficient. It does not claim that either method reproduces perception.

2.2.1 Scent Blending Mechanics

A paper released by Sisson et al. introduces a labeled dataset consisting of molecule pairs and compares each pair's catalog-recorded blend descriptors with the descriptors of its parent molecules using two baselines: union and intersection. The union score assumes that the blend record contains any descriptor listed for either parent (additive labeling). The intersection score assumes that the blend record contains only descriptors shared by both parents (subtractive labeling) [37].

The Jaccard results show 0.12 for union and 0.24 for intersection [37]. These low scores suggest that simple union or intersection rules do not reproduce the catalog-recorded labels of molecule pairs.

Catalog blend labels differ from the labels of the parent molecules in two ways. In catalog-label loss, a scent word recorded for either parent molecule is absent from the blend's labels. In catalog-label gain, the blend's labels contain a scent word that neither parent has. The results of Sisson et al. show that, on average, 1.2 notes are lost and 0.2 notes are gained per blend. Furthermore, an odds-ratio analysis reveals that certain scents, such as "eggy" or "cabbage", were more likely to be gained, while certain scents like "peach" or "fresh" were more likely to be lost [37]. These are differences between catalog records, not verified perceptual effects. They appear to follow a pattern rather than being random, which motivates testing a model that can learn relationships between the two molecules, such as a Set Transformer.

2.2.2 Evidence That Averaging May Be Sufficient

Not all evidence points to strongly non-additive mixing. Pellegrino et al. [28] had a trained panel rate 432 mixtures using a 51-term list and found that a linear average of the components' ratings predicted mixture profiles about as well as the noise in the panel's own ratings allowed. They argue that apparent emergence often reflects the limits of describing an odor with a few words. Binary catalog labels, as in Sisson et al. [37], instead record catalog-label loss and gain in many blends. Whether a learned pair-aggregation step helps for binary multi-label prediction is therefore an open question, which this study tests rather than assumes.

Other human studies describe more biology-focused ways in which a mixture can be perceived, such as odorants antagonizing one another [9, 13]. This study does not model receptors; the Set Transformer is evaluated only as a statistical aggregator of two learned vectors.

2.3 Machine Learning on Sets and Chemical Mixtures

This section is the core of the chapter. It defines permutation invariance, explains how mean pooling, order-averaged concatenation, and Set Transformer aggregation each combine two molecule vectors, and reviews prior work on mixture modeling.

2.3.1 Permutation Invariance and Deep Sets

A combination of two or more molecules could be viewed as a mathematical set. An accurate depiction of a fragrance blend of these molecules requires that they be presented in such a way that their positioning in that list is arbitrary, not sequential [37, 43, 52]. For instance, assume the combination of molecule A and molecule B produces scent C. It should be of no relevance whether it is molecule A + B or molecule B + A; it should always produce scent C. For that reason, it is important for any mathematical model used to predict the resultant smell from the combination of different molecules to follow this principle, which is called permutation invariance. In plain terms, a function is permutation invariant if swapping its inputs does not change its output.

Zaheer et al. [51] established the math of permutation invariance and introduced Deep Sets, a deep learning framework that uses a neural network designed to process datasets in a non-sequential manner for unordered data. This is referenced as the first of two properties they introduced in their study describing permutation invariance, positing that for a function acting upon a set, it must be indifferent to the order in which each element appears, mathematically formulated as $f : 2^X \to Y$ for any permutation, i.e., for any permutation π:

$$f(\{x_1, \ldots, x_M\}) = f(\{x_{\pi(1)}, \ldots, x_{\pi(M)}\}).$$

Their second result explains how to build such a function: transform each element individually (ϕ), aggregate the results, and then transform the aggregate again (ρ) [51].

However, because each element is transformed before pooling and in isolation, any information about how the elements relate to each other must be inferred indirectly by later layers from the pooled representation, as noted by Lee et al. [24]. For a pair of molecules, this raises the question of whether letting the two vectors affect each other before pooling helps.

2.3.2 Set Transformer (SAB and PMA)

To bridge this gap, Lee et al. propose the use of a Set Transformer, where instead of processing the elements of a set individually, they apply self-attention mechanisms they call SAB (self-attention block) in the processing of the elements, wherein the different elements are able to learn from each other, such that the extracted features to be used for pooling already contain useful information about the elements' relationship with each other. SAB itself is not permutationally invariant so swapping the order of the input gives different outputs. Moreover, they also developed PMA, or Pooling by Multihead Attention, which replaces the second step in the Deep Sets architecture wherein, instead of fixed sum or mean pooling, PMA also employs self-attention mechanisms in the pooling itself, such that it is able to pick out which of the elements bear more weight for the task at hand and gives a permutationally invariant output. Both are candidates for combining the two molecule vectors in this study: SAB lets each vector be updated using the other before pooling, and PMA learns how to weight them. The Set Transformer is therefore treated as a way to model statistical relationships between two learned vectors.

2.3.3 Three Ways to Combine Two Vectors

Three ways of combining the two vectors are compared in this study. Mean pooling averages the two vectors. It is simple and naturally order independent, but it can lose information specific to the pair, because different pairs with the same average produce the same combined representation [41]; this does not mean that it must fail on any particular task. Order-averaged concatenation places the two embeddings side by side in both possible orders, as in the pairwise design of Sisson et al. [37], and averages the two outputs. This adaptation makes the baseline invariant to input order while retaining both embeddings in full. Set Transformer aggregation lets the two vectors affect each other's learned representation before pooling, while remaining independent of order.

The three methods also differ in cost. Self-attention blocks in the Set Transformer scale quadratically with the number of elements in a set [24]; for a set of two molecules this cost is small, but attention blocks still add trainable parameters and computation compared with fixed pooling. A fair comparison therefore reports parameter counts and computation cost alongside accuracy, so that any gain can be weighed against the extra complexity.

2.3.4 Precedents in Mixture Modeling

Zaheer et al. [51] provided the groundwork for permutation invariance in sets, while Lee et al. [24] extended it with self-attention, a general set method that later studies applied to molecular mixtures.

Qin et al. [29] built SolvGNN to predict activity coefficients of binary and ternary chemical mixtures using two levels of graph processing: graph convolutions inside each individual molecule, followed by a molecule-level graph whose nodes are the molecules themselves. They argue that modeling the mixture explicitly at this second level matters in cases where naive pooling falls short [29].

Similarly, Sanchez Medina et al. [35] introduce GH-GNN, which also predicts activity coefficients of chemical mixtures. It uses a single shared molecular encoder whose pooled output is passed to a mixture-level graph. This resembles the structure of the present study, a shared single-molecule encoder followed by a mixture-level step, but applied here to odorant pairs rather than to solvent and solute.

Lastly, Tom et al. [43] proposed POMMix, a more relevant precedent for our topic, as it is in the field of olfactory perception, aiming to predict the similarity of the scent of two mixtures. It employs a module they call CHEMIX, which uses a self-attention mechanism, doing set-pooling to produce a permutation-invariant mixture embedding. POMMix differs from this study in its prediction task: it predicts the similarity between two mixtures, whereas this study predicts the scent labels of a two-molecule blend.

These three studies tackle different prediction problems. Evidence from solvent mixtures in particular does not transfer directly to scent labels. They motivate evaluating learned aggregation for odor blends; they do not establish that it outperforms simple pooling for the binary scent labels used here.

Hierarchical GNNs have also been used for property prediction within a single molecule, moving from atoms to fragments, as in HiGNN and HimGNN [17, 53]. POMMix, SolvGNN and GH-GNN, in contrast, operate across molecules. The present study follows the second kind: a shared single-molecule encoder followed by a pair-aggregation step for the two molecule embeddings.

2.4 Olfactory Datasets and Label Reliability

This section distinguishes the single-molecule catalog labels from the pairwise blend labels, describes what each can and cannot support, and explains why label inconsistency, label noise, and label sparsity matter for evaluation.

2.4.1 Reference Catalogs

The quality of sources for public scent data are not equal. Community platforms such as Fragrantica and Basenotes describe finished commercial perfumes in the words of their users, which makes them unsuitable for learning how individual chemicals relate to scent [7, 46]. Because of these, researchers use catalogs such as GoodScents and Leffingwell for reference, as they record descriptors for individual aroma chemicals and have been used to train state-of-the-art prediction models [23, 37]. The Leffingwell PMP 2001 database was compiled by olfactory experts, usually practicing perfumers, and its free-text descriptions were standardized into 113 descriptors by Sanchez-Lengeling et al.

2.4.2 The Blended-Pair Dataset

Sisson et al. [37] compiled blend labels from the lists of suggested blending partners on GoodScents, which group each partner under an odor heading. Their crawl was able to produce more than 160,000 pairs and, after standardization and filtering, resulted in up to 74 odor labels. Each molecule's page was crawled separately, so a pair can be listed from both molecules' pages. The dataset has no concentration information, and its labels are catalog suggestions and not measurements from an actual sensory panel. In this study they are therefore treated as catalog-recorded blend labels and used as proxy targets.

Note as well that the pair dataset used in this study is not assumed to be identical to the one used by Sisson et al., and that the pair scent labels are separate from the single-molecule labels in GoodScents and Leffingwell, which describe only single molecules.

Furthermore, a scent label may appear from one direction only (e.g., molecule A lists B as a potential blending partner for scent X, but B does not list A). Requiring both molecule pages to list each other for that scent label is one way to reduce reliance on a single recommendation. This study calls this two-page descriptor intersection, which is defined in Chapter 3. The trade-off is that it reduces the number of eligible labels, and it cannot show that a label is perceptually correct.

2.4.3 Human-Rated Datasets

Expert-compiled catalogs such as Leffingwell are assigned by people, but they are used here as training data and cannot serve as an independent check. Independent human-rated data are therefore needed. Human-rated scent datasets exist for single molecules and for mixtures, but to our knowledge none that gives scent descriptors for two-molecule blends is publicly available as of November 2026. Bierling et al. [6] provide yes-or-no ratings of 16 scent words for 74 single odors from about 1,200 laypeople. Pellegrino et al. [28] collected trained-panel ratings of mixtures with a 51-term list.

2.4.4 Absence of a Standard Vocabulary

No standard set of odor descriptors exists. Catalogs and studies use vocabularies of different sizes, and many words overlap in meaning or differ only in spelling, such as “acid” and “acidic” [37]. Sajan et al. [34] grouped raw odor words from the Pyrfume repository into 16 broader scent families, but this study does not use those families because their words are not directly equivalent to the descriptors in the present data. The models therefore predict only the standardized descriptors defined for this dataset. How the vocabulary is standardized and filtered, and how unlisted descriptors are treated, is described in Chapter 4.

2.4.5 Noisy and Imbalanced Labels in Multi-Label Learning

Inconsistent labels act as noise in both the training targets and the test targets. Noise in the training targets can change what a model learns, because deep networks can memorize noisy labels [38]. Noise in the test targets can change which model appears best. Noise is harder to handle with several labels per example, because errors can be correlated across labels; Li et al. [25] model this by estimating a noise transition matrix that uses label correlations. Moreover, missing scent labels is an additional problem because unlisted scent labels may not necessarily be true absence and were just omitted by the catalog. For this reason, this study measures label consistency, evaluates on the most consistent labels, and tests whether the comparison between aggregation methods changes as label reliability decreases.

Multi-label scent data are also sparse: most descriptors are absent for most molecules, and some descriptors are far more common than others. Reviews of imbalanced multi-label classification describe imbalance within labels, among labels, and among label sets [42], and work on graph neural

networks reports that the benefit of balancing techniques depends on the dataset and problem type [2]. Because simple accuracy-style scores can look good on sparse labels, evaluation of such data needs metrics that are not dominated by true negatives [30].

2.5 Out-of-Distribution Splitting Strategies

This section provides a review of the standard data-splitting strategy of scaffold splitting, and an explanation of how its application could fall short in sparse chemical spaces. This section then grounds and discusses the clustering-based alternative used in this study.

A common method to partition data for training, testing, and validation in chemoinformatics and molecular machine learning is scaffold splitting [15, 39]. In this method, molecules are represented by graphs of their structure, and the core framework of these graphs is referred to as their "scaffold" (Bemis–Murcko scaffolds) [48]. Molecules that share the same scaffold are grouped together, and these groups are then distributed among the training, testing, and validation sets to ensure that the test set contains structurally different molecules from the training set, which is a basic form of out-of-distribution (OOD) evaluation, which as opposed to random splitting, simply distributes molecules randomly across training, validation, and test sets [12, 15, 37, 39].

The caveat with this method is that, when the dataset is sparse, it collapses toward random splitting. For instance, similarly structured molecules could get fragmented across both the train and test splits, leading to data leakage and inflated test scores as scaffolds in standard benchmarks could be populated by only one or two molecules each, undermining the grouping effect the method depends on [12]. Another problem demonstrated by scaffold splitting is that even molecules that are near-identical (as computed by Tanimoto similarity) [15, 39], even differing by as little as a single atom, can, under this method, end up in opposite splits, as observed by Guo et al. and Steshin et al. [15, 39], who argue that this problem causes models to "memorize" rather than generalize from the data. Because of these problems, scaffold splitting can misleadingly produce scores that are overly optimistic.

Shimakawa et al. [36] propose doing topological pre-clustering by projecting 2048-bit ECFP4 fingerprints onto a 2D map using 2D Uniform Manifold Approximation and Projection (UMAP), then grouping them together using spectral k-means clustering. More generally, leakage-reduced splitting can be formulated as an optimization problem over the data [22], and cluster-based splits

have been reported to be among the most challenging for both classical and graph-based models [12].

In plain terms, one molecule can appear in many blend pairs, so splitting pairs at random can place the same molecule in both training and testing. Methods designed for single molecules do not address this, because an individual molecule can be a component of varying other molecules in different blends, meaning a molecule seen during training could reappear as a component in a different, unseen test blend, leaking information about that molecule across the split and inflating the performance metrics. Therefore, Sisson et al. [37] propose "graph carving", using a meta-graph to describe the pairwise blend dataset, where each node represents an individual molecular graph and each edge represents a tested blend combination. Edges connecting a molecule from one partition to a molecule in another are discarded, so that no component is shared between splits. In their original implementation, this partitioning was done by randomly assigning molecules to each side and repeating until every label had at least one training and one test example, since a true scaffold-based split was not compatible with their sparse label distribution.

This study applies graph-carving protocol on top of a fingerprint-clustering partition (in the style of the UMAP and k-means approach discussed above), rather than random or scaffold-based partitioning. Doing this creates a component-disjoint split that reduces component and structural overlap between training and testing; it does not guarantee that every test molecule is entirely unlike every training molecule.

2.6 Evaluating and Comparing Models Under Label Sparsity

Sparse multi-label targets make some common scores misleading. Precision, recall, and F1 ignore true negatives and so expose a model that predicts no labels, whereas accuracy-style scores such as Hamming loss can look strong for such a model [30]. F1 can be averaged per descriptor (macro), which gives scent labels that are rarely found the same weight as common ones, or pooled across all decisions (micro), which favors common scent labels [30]. Average precision (area under the precision–recall curve) summarizes how well a model ranks true scent labels above false ones across all cutoffs. The cutoff used to turn a probability into a present or absent prediction is also important. A study comparing graph-based and traditional models for multi-label odor prediction found that choosing the right threshold was essential [45]. To ensure fair results, this threshold should be

selected without using the test set.

Comparisons between models should also report how large a difference is. Bootstrap confidence intervals do this by showing the difference and its uncertainty together [5]. Recent work warns against naive paired t-tests when resampled test sets are used [30]. Together with the leakage problem above, this means that a fair comparison of aggregation methods needs matched encoders and training conditions, sparsity-aware metrics, validation-only threshold selection, and paired uncertainty estimates over the evaluated blends.

2.7 Synthesis and Research Gap

2.7.1 Synthesis of Literature

Current models in computational olfaction can successfully map single-molecule structures to their appropriate sensory descriptors [23]. However, because these compounds are evaluated in isolation, they do not address how two molecules should be combined, and catalog blend labels differ from the labels of the parent molecules in many blends [37]. Human-panel evidence on mixtures is mixed: continuous ratings are well predicted by a linear average of the components [28], while binary catalog labels record catalog-label loss and gain in many blends [37].

From an architectural standpoint, order-independent set processing is a natural requirement for combining unordered components [51, 24]. Averaging can make different inputs look identical, and a related loss of information has been described for averaging-based readouts within a molecule [41]. For a pair, this is a possible limitation of mean pooling.

Furthermore, standard scaffold splits can behave like a random split and can cause data leakage [12]. Clustering molecules and carving the blend meta-graph along cluster boundaries reduces component and structural overlap between training and testing [36, 37]. Finally, the blend labels in public data come from catalogs rather than sensory panels, and their reliability has rarely been measured in prior blend studies.

In summary: (1) a molecular graph encoder produces one vector per molecule; (2) the unresolved computing problem is how to combine the two vectors without making the prediction depend on input order; and (3) any such comparison must account for component leakage and uncertain labels.

2.7.2 Statement of the Research Gap

To our knowledge, existing work does not yet establish whether an attention-based set aggregator provides a reliable advantage over simpler alternatives when two molecular graph embeddings are combined for sparse, catalog-derived multi-label prediction. A fair answer requires the encoder and training conditions to be controlled, input order to be handled explicitly, repeated molecular components to be separated across partitions, and uncertain labels to be considered during evaluation. This study addresses that controlled evaluation gap.

2.7.3 Positioning of the Present Study

To address this gap, this study compares mean pooling, order-averaged concatenation, and Set Transformer aggregation for combining the two molecule embeddings produced by one shared, frozen graph attention encoder, with a separately trained prediction head for each method. The comparison is carried out on a component-disjoint, fingerprint-clustered and graph-carved partition of the data. The two-page descriptor intersection screen and the controlled label-corruption tests show how far the conclusions depend on the labels. A simpler baseline that matches or outperforms the Set Transformer would be a valid and informative result.

Chapter 3

Theoretical and Conceptual Framework

3.1 Theoretical Framework

This chapter follows the computation from the molecular input to the final evaluation. It first explains how a molecule is represented as a graph of atoms and bonds. It then describes how message passing and graph attention produce one embedding for each molecule. The next part explains why combining two embeddings is a separate design problem: a blend is an unordered pair, so the prediction should not depend on which molecule is listed first. The three aggregation methods are then described and compared. The chapter ends by explaining why sparse, catalog-derived labels affect both training and evaluation, and how reciprocal agreement, cross-catalog agreement, and controlled label corruption are used to study that problem. The discussion of averaging describes a possible information loss, not an observed result; the relative performance of the methods remains an empirical question.

3.1.1 Representing Molecules as Graphs

Each molecule is represented as an undirected, attributed graph. Atoms are the nodes, bonds are the edges, and the features attached to them become the inputs to the graph encoder [49]. This representation keeps the bond connectivity of the molecule instead of treating the structure as an

ordinary sequence of characters. SMILES strings are used to store and parse the structures, but different valid atom-numbering or starting-point conventions should describe the same molecular graph [27, 46].

Mathematical Formalism

Each molecule is defined as an undirected, attributed graph $G = (V, E)$, where V is the set of non-hydrogen atoms and E is the set of covalent bonds. In the representation used here, hydrogen information is included in the features of the corresponding heavy atoms rather than represented as separate nodes [4].

The connectivity can be written as an adjacency matrix A, where $A_{ij} = 1$ when atoms i and j are connected by a bond and $A_{ij} = 0$ otherwise [53]. The implementation stores this connectivity in a sparse form (Section 3.3.2). Each node and edge also has a feature vector:

- The node vector $\mathbf{x}_i \in \mathbb{R}^{d_x}$ records properties such as element, charge, valence, hybridization, aromaticity, hydrogen count, ring information, and chirality.
- The edge vector $\mathbf{e}_{ij} \in \mathbb{R}^{d_e}$ records bond properties such as bond type, conjugation, ring membership, and stereochemical configuration.

The complete node and edge feature tables are provided in Appendix A. RDKit supplies the parsed molecular structures and their features. The graph representation is not tied to a particular atom numbering: relabeling the nodes changes their order in the data structure, but not the bonds or the molecule they describe. This property is important because the message-passing and pooling operations should produce the same molecule representation when the atoms are listed in a different order. In this study, $d_x = 46$ and $d_e = 10$, with the components listed in Appendix A.

3.1.2 Graph Neural Networks and Message Passing

Most neural networks receive inputs in a fixed arrangement. A molecular graph does not have that form: molecules can contain different numbers of atoms, and their important relationships are represented by bonds. A Graph Neural Network (GNN) works with this structure directly. In the graph used here, a *node* represents an atom and its feature vector, while an *edge* represents a bond and its feature vector. The *neighborhood* $\mathcal{N}(i)$ of node v_i is the set of atoms directly connected to it by an edge [53, 4, 45].

The main operation in a GNN is *message passing*. During one layer, each node receives information from its neighborhood, combines those messages, and updates its own representation. For node v_i at layer t, one general form of this operation is:

$$\begin{aligned} \mathbf{m}_i^{(t)} &= \bigoplus_{j \in \mathcal{N}(i)} M_t\big(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}\big), \\ \mathbf{h}_i^{(t+1)} &= U_t\big(\mathbf{h}_i^{(t)}, \mathbf{m}_i^{(t)}\big). \end{aligned} \tag{3.1}$$

Here, $\mathbf{h}_i^{(t)}$ is the current representation of node i, $\mathbf{e}_{ij}$ describes the bond to neighbor j, and M_t computes the message sent along that bond. The symbol $\bigoplus$ denotes the operation that combines all messages received by node i, while U_t produces its new representation. In ordinary terms, the node looks at itself, looks at its bonded neighbors, and then records an updated summary. Since the messages are combined as a group rather than in a prescribed order, changing the order in which neighboring atoms are stored does not change the information being combined. Repeating the layers allows information to travel from an atom to nearby atoms and then to larger parts of the molecule [17, 53].

After the final message-passing layer, a *graph readout* combines the node representations into one vector for the whole molecule. The readout specified for the final protocol is applied consistently to both molecules; when mean pooling is used, the node vectors are averaged before being passed to the blend-level models. This node-to-molecule step is common in molecular GNNs, although some recent models also learn representations for larger substructures [17, 45].

The GNN used here is a Graph Attention Network (GAT). Rather than assigning the same fixed weight to every neighboring node, a GAT learns weights from the node and bond features, allowing the encoder to give more influence to some local connections than to others. These weights are model parameters used to form a representation for descriptor prediction; they do not establish a receptor mechanism or a physical interaction between molecules [4, 45]. The specific attention calculation is described in the next subsection.

3.1.3 Graph Attention Encoder

The shared graph encoder maps each molecular graph to a fixed-length vector that can be used by any of the three blend-level aggregation methods. It learns representations of local atom and bond context that are useful for the descriptor prediction task. The same encoder is applied to the two molecules in a blend, so the resulting vectors are produced under the same learned rules [23, 49].

Let $\mathbf{h}_i^{(0)} = \mathbf{x}_i$ denote the 46-dimensional initial feature vector of node v_i, and let $\mathbf{e}_{ij}$ denote the 10-dimensional edge feature vector. The approved encoder in Chapter 4 uses three graph-attention layers and produces a 256-dimensional molecule embedding. A learnable matrix projects the node states into the hidden space. To score how much node v_i should attend to neighbor v_j, a small learnable scoring function uses the projected states of both nodes and the edge features [49, 53]:

$$\tilde{e}_{ij} = \mathrm{LeakyReLU}\Big(\mathbf{a}^T\big[\mathbf{W}\mathbf{h}_i \,\|\, \mathbf{W}\mathbf{h}_j \,\|\, \mathbf{W}_e\mathbf{e}_{ij}\big]\Big), \tag{3.2}$$

where $\|$ denotes concatenation. Including $\mathbf{e}_{ij}$ lets the score use information about the bond connecting the two nodes. The activation and its parameters are part of the approved encoder specification [49].

Atoms have different numbers of neighbors, so the raw scores are normalized across the neighborhood $\mathcal{N}(i)$ of atom v_i with a softmax, so that the weights for each atom sum to one [49, 4]:

$$\alpha_{ij} = \mathrm{Softmax}_j(\tilde{e}_{ij}) = \frac{\exp(\tilde{e}_{ij})}{\sum_{k\in\mathcal{N}(i)}\exp(\tilde{e}_{ik})}. \tag{3.3}$$

The normalized weights α_{ij} scale the projected neighbor features, and the selected node-update activation produces the updated representation of node v_i [49]:

$$\mathbf{h}_i^{(l+1)} = \sigma_{\mathrm{upd}}\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}\mathbf{W}\mathbf{h}_j^{(l)}\right). \tag{3.4}$$

Here σ_{upd} denotes the node-update activation selected in the final encoder specification.

The encoder repeats this update across three layers. The first two use four attention heads, and the last produces the final 256-dimensional node states. A learned global-attention readout then combines the final node states into one molecule vector $\mathbf{z}$ passed to the blend-level model. Thus, the node states retain local and progressively broader graph information, while the readout makes that information available as one vector for the molecule.

The encoder is pretrained once on the training-partition single molecules and then frozen for the primary comparison. The three aggregation variants therefore receive embeddings produced by the same encoder weights and use matched training schedules, optimization settings, and budgets.

The attention coefficients are learned computational weights used to form the representation. They indicate which neighboring features the model emphasizes for its prediction, but they do not demonstrate chemical importance, receptor activity, or a physical interaction between molecules.

3.1.4 Combining an Unordered Pair

The blend model receives two molecule embeddings, $\mathbf{z}_A$ and $\mathbf{z}_B$. Because the pair has no meaningful first or second member, the model should return the same prediction for (A, B) and (B, A). In other words, changing the order in which the two molecules are supplied must not change the predicted descriptor vector [37, 51].

The three aggregation methods satisfy this requirement in different ways. Their pair representations and main trade-offs are summarized below.

Table 3.1: Aggregation methods for an unordered pair of molecule embeddings.

Method	**Pair representation**	**How order is handled**	**Main trade-off**
Mean pooling	Average the two embeddings	Invariant by construction	Can discard pair-specific distinctions
Order-averaged concatenation	Evaluate both concatenation orders and average their predictions	Explicitly symmetrized	Requires two ordered evaluations
Set Transformer aggregation	Let the two embeddings exchange learned information before pooling	Invariant set processing	Adds parameters and computation

Mean Pooling

Mean pooling is the simplest order-independent baseline. It represents the pair by the element-wise average

$$\mathbf{z}_{\text{mean}} = \frac{\mathbf{z}_A + \mathbf{z}_B}{2}. \tag{3.5}$$

Swapping the two inputs gives the same vector automatically, so this method is invariant by construction. Its limitation is that different pairs can share the same average. For example, the one-dimensional pairs $(2, 8)$ and $(4, 6)$ both produce 5. This possible loss of pair-specific distinction motivates the comparison with the other two methods; it is not a claim about the observed results [51, 41].

Order-Averaged Concatenation

Concatenation keeps the two embeddings side by side, but one ordered concatenation depends on which molecule is listed first. The adapted baseline therefore evaluates both $[\mathbf{z}_A \,|\, \mathbf{z}_B]$ and $[\mathbf{z}_B \,|\, \mathbf{z}_A]$,

then averages the two resulting model outputs. This explicit symmetrization makes the final prediction the same for both input orders while retaining more pair-specific information than a simple average. It requires two ordered evaluations for each pair.

The two logits are averaged before a single sigmoid is applied. This is an adapted baseline based on the pairwise concatenation design of Sisson et al., not a reproduction of their complete model [37].

Set Transformers and Attention-Based Pooling

The Set Transformer receives the two embeddings as a set $X = \{\mathbf{z}_A, \mathbf{z}_B\}$. Its attention blocks allow the embeddings to exchange learned information before a pooling step produces one blend representation [24, 43]. The Self-Attention Block (SAB) processes the two members without assigning them a permanent first or second position, so the resulting set representation is unchanged when the inputs are swapped. The method can represent statistical relationships between the learned embeddings that mean pooling may discard, but it adds parameters and computation. The approved configuration uses two SAB blocks with four attention heads and a 512-unit row-wise feedforward layer.

The second block is Pooling by Multihead Attention (PMA):

$$\mathrm{PMA}_k(X) = \mathrm{MAB}\big(S, \mathrm{rFF}(X)\big). \tag{3.6}$$

This process begins by passing the molecular embeddings $\mathbf{X}$ through a row-wise feedforward transformation. The PMA block uses learned seed vectors S to attend to these transformed embeddings and compress them into a fixed-size blend representation [24, 37]. In this study, PMA uses one learned seed vector, and its attention weights are interpreted as computational parameters for combining representations. They model statistical relationships between embeddings for descriptor prediction, not physical molecular interactions or receptor mechanisms. Whether the additional capacity improves prediction is tested empirically under the same training and evaluation conditions as the baselines.

3.1.5 Representation Trade-offs

The three methods make different compromises when two molecule embeddings are reduced to one blend representation. Mean pooling is inexpensive and order independent, but different pairs

can have the same average, so some pair-specific distinctions may be lost [51, 41]. It remains a credible baseline because its simplicity can be sufficient when the averaged embedding retains the information needed for the descriptor task. The order-averaged concatenation baseline retains the two embeddings side by side and restores order invariance by evaluating both input orders, at the cost of a second ordered evaluation. Set Transformer aggregation has additional learned capacity and computation, and uses that capacity to model statistical relationships between the embeddings before pooling. These are representational trade-offs, not assumptions about the chemistry of a blend.

Why Averaging Can Lose Information

Averaging two molecule embeddings can make structurally different pairs indistinguishable to the model. In one dimension, the pairs $(2, 8)$ and $(5, 5)$ both average to 5, so the averaged representation cannot distinguish those pairs. This example demonstrates a possible information loss; it does not show that such collisions are common or dominant in the present dataset. The actual usefulness of mean pooling remains an empirical question. The order-averaged concatenation baseline retains the two embeddings side by side while correcting for the two possible input orders, and the Set Transformer provides another order-invariant way to combine them by learning statistical relationships before pooling. A full derivation of the symmetry argument is given in Appendix A.1.

3.1.6 Sparse Labels and Label Reliability

The output is a sparse multi-label vector: most blend–descriptor cells are absent, while a smaller number are recorded as present. This imbalance affects both parts of the experiment. During training, the many easy negative cells can dominate the loss. During evaluation, a model that predicts mostly negatives can look favorable on an aggregate score even when it misses useful positive descriptors. The study therefore uses metrics suited to sparse multi-label outputs and applies a reliability mask when only the more consistent cells are being evaluated. The exact mask construction and training conditions are specified in Chapter 4.

Because the blend labels used in this study are compiled from catalog recommendations rather than a controlled sensory panel, their reliability must be considered before model comparisons can be interpreted. The reliability checks describe the strength and consistency of the catalog evidence; they do not establish perceptual truth.

Reciprocal agreement is a within-catalog check. In the GoodScents catalog, molecule A's page may list molecule B as a blending partner under a given scent word, but molecule B's page may not list molecule A under the same word. When both pages agree, the label is *reciprocal*; when only one page makes the claim, the evidence is weaker. Reciprocal evidence therefore strengthens confidence in a recorded positive label, but it does not prove that the descriptor is perceptually correct.

Cross-catalog agreement is measured using Cohen's kappa (κ), which quantifies how consistently two independently maintained catalogs (GoodScents and Leffingwell) assign the same descriptor to the same molecule, correcting for the agreement expected by chance alone. A commonly used benchmark scale classifies κ values between 0.21 and 0.40 as "fair" agreement, 0.41 to 0.60 as "moderate," and above 0.60 as "substantial" agreement [44]. These categories are used as operational descriptions of descriptor-level consistency, not as universal or objective judgments of label correctness. The thresholds and the catalogue records used to compute them are specified in Chapter 4.

The target representation follows a closed-world assumption for the purpose of the supervised task: an unlisted descriptor is encoded as a negative target. This is a modeling convention, not a claim that the odor is truly absent; catalogs can omit descriptors for reasons unrelated to sensory absence.

Reliability is applied at the blend–descriptor cell level. An eligibility-mask comparison must include eligible negative cells as well as reliable positive cells. Otherwise, a test set containing only recorded positives could not penalize false-positive predictions, making measures such as F1 or PR-AUC misleading. Cells whose evidence is ambiguous are masked from the relevant loss or metric rather than causing the entire blend to be discarded. Weaker evidence does not prove that a catalog label is wrong.

Even after reliability screening, some label noise may remain. The controlled label-corruption experiment described in Chapter 4 tests how the aggregation methods respond as training-label reliability is reduced. A smaller performance decline under the same corruption condition would indicate greater robustness to that tested form of noise, but it would not establish that the labels are perceptually accurate [38].

3.2 Conceptual Framework

The conceptual framework of this study operationalizes the theoretical formulations into a systematic computational pipeline structured around the Input-Process-Output (IPO) model. Figure 3.1 outlines the structural interaction between input data representations, the two-level hierarchical deep learning architecture, and the downstream analytical and practical deliverables.

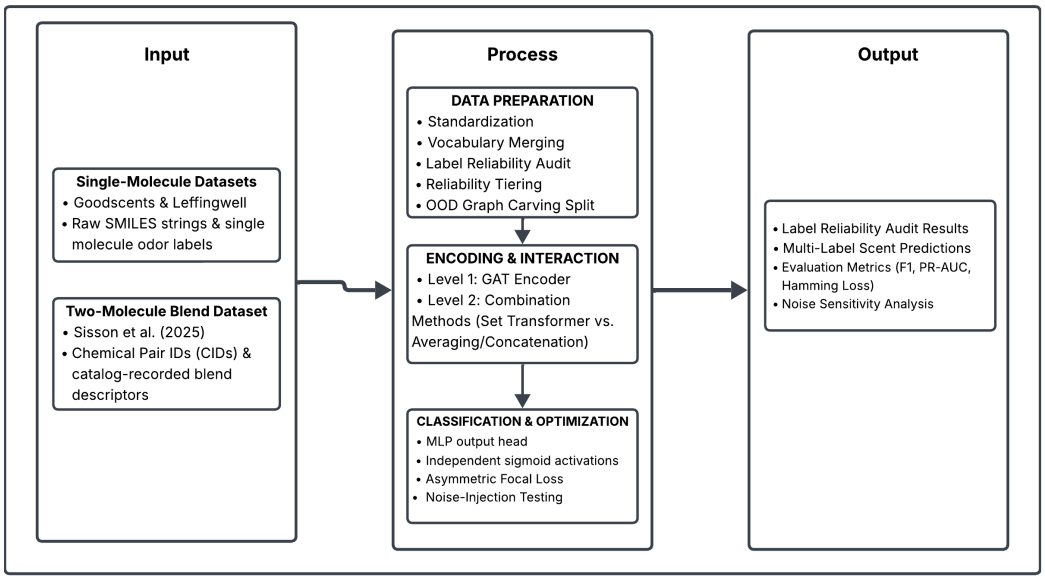


Figure 3.1: Input-Process-Output (IPO) Model.

3.2.1 Input Phase

The computational data pipeline starts with the acquisition, curation, and gathering of data from olfactory databases. The raw SMILES strings are extracted from the GoodScents and Leffingwell repositories accessed via the Pyrfume Library in Python [7]. These raw chemical blueprints are then paired with catalog-recorded odor tags as proxy targets for pretraining. The binary mixture profiles are retrieved from the Sisson et al. (2025) benchmark dataset [37]. The retrieved benchmark dataset provides standardized chemical pair identifiers (CIDs) and catalog-recorded descriptors that will be used for multi-label prediction of two-molecule blends.

3.2.2 Process Phase 1: Data Preparation and Graph Carving

To raw chemical input will first pass through a standardization pipeline. Salts and other non-fragrance components are removed using RDKit's SaltRemover module, leaving the main molecule. Compounds containing elements other than C, H, O, N, S, F, Cl, Br, and I are dropped, since these fall outside the range typical of aroma chemicals [27]. The molecular weight should be between 20 and 600 g/mol. Below this range is considered irrelevant, while above the range is considered too heavy to be perceived as a scent at all [23]. The surviving SMILES strings are canonicalized using RDKit. The scent descriptors text from the source databases is standardized by merging redundant terms and eliminating descriptors that appear rarely to learn from [8]. The blend labels are then audited for reliability before any model is trained. Reciprocal agreement is measured for each blend pair, and cross-catalog agreement between GoodScents and Leffingwell is measured on the parent molecules using Cohen's kappa. The resulting evidence is used to construct the cell-level reliability conditions described in Chapter 4. These conditions describe stronger or weaker catalog evidence; they do not prove that a label is true or false. Unlisted descriptors are treated as negative targets under the study's closed-world assumption, while ambiguous cells are masked rather than used to remove an entire blend. To test how well the network handles unfamiliar compounds, molecules are grouped into structural families by the approved clustering procedure. The Graph-Carving algorithm then divides the mixture meta-graph along family clusters. The resulting component-disjoint procedure reduces overlap of molecular components between partitions; it does not guarantee that held-out molecules are structurally unrelated to training molecules [12].

3.2.3 Process Phase 2: Encoding and Pair Representation

The second phase converts each molecular graph into a fixed-length vector. Atoms and bonds are processed as graph nodes and edges, and the shared GAT encoder updates the node representations before applying global-attention readout [49]. The same frozen encoder is used for both molecules and all three aggregation variants. The dimensions and layer settings follow the fixed specification in Chapter 4.

The two molecule embeddings are then combined into a single blend representation using one of three aggregation methods: mean pooling, order-averaged concatenation following the adapted pairwise baseline of Sisson et al. [37], or Set Transformer aggregation. These methods are compared

under matched training schedules. The Set Transformer is treated as a learned way to combine the two representations; the study does not claim that its attention weights describe physical or receptor-level interactions.

3.2.4 Process Phase 3: Classification and Optimization

Once the aggregation module yields the blend representation, it is translated into multi-label descriptor predictions. The Multi-Layer Perceptron (MLP) head maps the 256-dimensional blend representation through 128 hidden units and ends with L individual outputs to match the target scent classes [37]. Because scent profiles are inherently multi-label, the classification layer utilizes independent sigmoid activation functions [33]. These activations compute individual probability scores for the joint presence or absence of each unique target scent descriptor. Because the encoder is frozen for the primary comparison, prediction errors update only the aggregation and output modules. The loss uses the fixed imbalance parameters stated in Chapter 4 [18]. Each combination method is trained under the reliability conditions specified in Chapter 4, including a condition that uses eligible high-reliability cells and their eligible negative cells. The noise-sensitivity test applies the declared controlled label corruption to the same matched conditions. Olfactory datasets are highly sparse, meaning a single scent profile contains only a few active descriptors alongside dozens of inactive ones (zeros). Under standard training rules, the model could appear accurate simply by predicting "no" for most labels. The asymmetric loss addresses this class imbalance by reducing the influence of obvious negatives and forcing the network's mathematical focus onto the rare, active scent notes that are difficult to predict [18].

3.2.5 Output Phase

The final phase outputs a multi-label binary vector of predictions across all L target scent descriptors which is based on the probability given to each descriptor. A validation module evaluates the model's performance on these predictions under the declared reliability masks. It calculates metrics such as macro- and micro-averaged F1-scores, Hamming loss, and the Precision-Recall Area Under the Curve (PR-AUC) [3, 42, 45]. The reliability-aware comparison includes eligible negative cells so that false-positive predictions remain evaluable; ambiguous cells are excluded by the mask. The exact partitions, masks, and statistical comparison procedure are specified in Chapter 4.

The study additionally reports the results of the label reliability audit, including reciprocal agreement rates and cross-catalog agreement by descriptor.

3.3 Technical Background

Mapping molecular structure onto multi-label olfactory predictions requires stitching together a few different layers of infrastructure: cheminformatics standards to make sure the molecules are valid, a graph learning framework to process them, and a loss function suited to how sparse the labels are. This section walks through each piece—the parsing standards, the sparse graph representation, and the loss objective—and explains why each choice was necessary rather than incidental.

3.3.1 Cheminformatics Standard (RDKit): SMILES Parsing and Sanitization

Molecular structures in this study are represented as Simplified Molecular Input Line Entry System (SMILES) strings, a compact text-based notation, which are canonicalized and parsed into undirected, attributed molecular graphs using the open-source RDKit library in Python [27]. SMILES strings are convenient because they are compact and suitable for storage. This compactness comes at a cost, however: they are structurally sensitive, prone to syntax errors, and do not encode explicit 3D geometry [46]. Before any structure reaches the model, every string passes through RDKit's sanitization pipeline, which runs four checks in sequence:

- Kekulization checks that the alternating single and double bonds within conjugated or aromatic rings are represented in a chemically consistent manner, resolving ambiguous resonance structures.
- Aromaticity Perception evaluates the coordinate and topological structure of each atom to identify aromatic rings, dynamically reassigning aromatic bond attributes to relevant edges.
- Valence Checking checks every atom, specifically heavy atoms–carbon, nitrogen, oxygen, sulfur, and halogen to ensure it does not exceed its physical valency limits. This rejects syntactically corrupted SMILES strings or physically impossible hypervalent structures.
- Ring Perception, which identifies and indexes ring systems, information the later convolutions depend on.

Enforcing these rules ensures that what the network receives is already known to be chemically valid, rather than requiring the model to infer the rules of organic valency implicitly from training data [4, 53]. The sanitization pipeline acts as an explicit structural filter ahead of the model, so that the Level 1 Graph Attention Network (GAT) operates only on topologically valid graphs that correspond to real molecular structures [23, 49].

3.3.2 Deep Learning Stack (PyTorch / PyTorch Geometric): Sparse Graph Convolutions

After sanitization, each molecule is formatted as an undirected graph $G = (V, E)$, where atoms are nodes (V) and covalent bonds are edges (E). A dense $N \times N$ adjacency matrix could represent this connectivity, but for molecules, where most atoms bond to only a few neighbors, this wastes memory, scaling at $\mathcal{O}(N^2)$ for structures that are inherently sparse. The architecture instead runs on PyTorch Geometric, which handles molecular graphs as sparse attributed objects rather than dense arrays [11].

Topology is stored in Coordinate (COO) format: a sparse two-row index matrix C of size $2 \times 2E$, where the first row lists source node indices and the second lists receiver node indices [4]. Since chemical bonds are physical and undirected, every source node is also a receiver, so $2E$ entries are needed rather than E — each bond is represented in both directions so that message-passing can update both of its atoms. Node states are updated by running the GAT computation described in Section 3.1.3 over this sparse index matrix, rather than over a dense adjacency matrix, at each layer l.

3.3.3 Multi-Label Mathematical Formulation: Independent Sigmoids and Sparsity Mitigation

A molecule or blend can carry several scent descriptors at once (for example, both fruity and green), rather than belonging to a single exclusive category. The descriptors can therefore co-occur, and predicting one descriptor should not force all other descriptors to be absent. The final layer skips a mutually exclusive softmax and instead maps the approved blend embedding through an MLP to L independent sigmoid outputs, one per target descriptor [33]. Each output is a probability $\hat{y}_{ik}$

between 0 and 1 for example i and descriptor k:

$$\hat{y}_{ik} = \sigma(s_{ik}) = \frac{1}{1 + e^{-s_{ik}}}. \tag{3.7}$$

Here s_{ik} is the output score before the sigmoid. The target y_{ik} is binary for each descriptor, so the model makes one present/absent decision per descriptor rather than one choice from a mutually exclusive list.

A multi-label dataset is sparse when the number of positive cells is small relative to the total number of example–descriptor cells. Let N be the number of examples and L the target-vocabulary size. The label density can be described without fixing a provisional value as $\rho = (NL)^{-1} \sum_{i=1}^{N} \sum_{k=1}^{L} y_{ik}$. A small ρ means that most cells are negative. The many easy negatives can then dominate a plain loss, allowing a model that predicts mostly zeros to appear successful while missing the positive descriptors that matter for the comparison.

A standard multi-label setup would optimize independent Binary Cross-Entropy terms across the L descriptors:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{NL} \sum_{i=1}^{N} \sum_{k=1}^{L} \Big[y_{ik} \log(\hat{y}_{ik}) + (1 - y_{ik}) \log(1 - \hat{y}_{ik}) \Big]. \tag{3.8}$$

The planned loss is an Asymmetric Focal Loss. It separates the focusing of positive and negative terms through exponents γ_p and γ_n and uses a class-frequency weight α_k [18]:

$$\mathcal{L}_{\mathrm{FL}} = -\frac{1}{NL} \sum_{i=1}^{N} \sum_{k=1}^{L} \Big[y_{ik} \alpha_k (1 - \hat{y}_{ik})^{\gamma_p} \log(\hat{y}_{ik}) + (1 - y_{ik})(1 - \alpha_k) \hat{y}_{ik}^{\gamma_n} \log(1 - \hat{y}_{ik}) \Big]. \tag{3.9}$$

The exponents and class weights determine how strongly easy negative terms are down-weighted relative to rarer positive terms. Chapter 4 fixes $\gamma_p = 0$ and $\gamma_n = 2$ for the comparison. The purpose of the loss is to reduce the effect of the imbalance; it does not remove the need for reliability masks or appropriate evaluation metrics.

3.3.4 Evaluation Metrics for Sparse Multi-Label Outputs

Macro F1 is the primary metric for the comparison. It computes F1 separately for each evaluated descriptor and then averages those values, so frequent descriptors do not determine the result by themselves. This is useful when the study compares performance across a vocabulary whose descriptor frequencies may differ.

Micro F1 is reported as a supporting metric by pooling true positives, false positives, and false negatives across descriptors before calculating one F1 value. PR-AUC is also supporting: it measures how well the model ranks positive cells across decision thresholds, which is informative when positives are rare. Hamming loss reports the fraction of evaluated cells classified incorrectly, so it gives a complementary view of overall cell-level errors. The metric definitions and masking rules are specified in Chapter 4.

Hamming loss must be interpreted against an all-negative reference because a sparse label matrix can make an always-negative predictor appear strong. Let M_{ik} be the evaluation mask and define $\hat{y}_{ik}^{(0)} = 0$ for every eligible cell. The corresponding reference is

$$\mathrm{HL}_{\mathrm{neg}} = \frac{\sum_{i=1}^{N} \sum_{k=1}^{L} M_{ik} y_{ik}}{\sum_{i=1}^{N} \sum_{k=1}^{L} M_{ik}}. \tag{3.10}$$

This is the error made by predicting negative for every evaluated cell. A model's Hamming loss should therefore be reported with this reference rather than interpreted in isolation. The reference is computed separately for each declared evaluation mask.

Chapter 4

Methodology

This chapter describes how the catalog-derived data were prepared, how the label reliability masks were built, how component-disjoint partitions were formed, and how three aggregation methods were trained and evaluated under matched conditions. It also defines the robustness conditions, metrics, and paired comparisons used to answer the research questions.

4.1 Research Design

This study uses a quantitative experimental design on a multi-label molecular graph classification task. It compares three ways of combining the learned embeddings of the two molecules in a blend: mean pooling, order-averaged concatenation, and Set Transformer aggregation. Experiments A and B provide the matched aggregation comparison for RQ1, the blend-type analysis addresses RQ2, and controlled label corruption addresses RQ3.

Table 4.1 states what is varied and what is held fixed. The aggregation method is the primary varied factor. The label condition and the corruption level are robustness factors, and the catalog-label pattern of a blend is used only for a subgroup analysis.

Figure 4.1 gives an overview of the implemented pipeline.

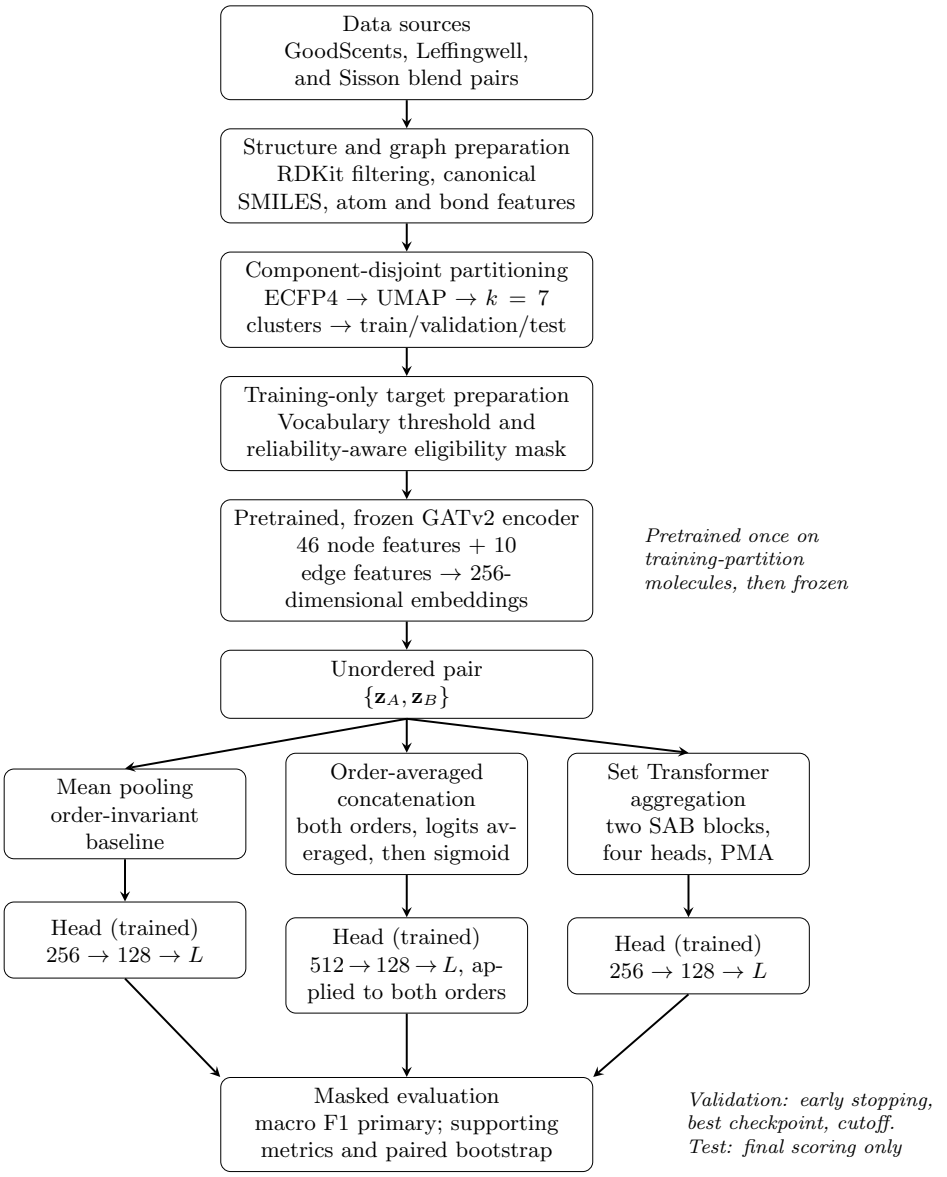


Figure 4.1: Overview of the implemented pipeline. The three aggregation methods receive the same frozen embeddings and the same evaluation protocol, and each has its own separately trained prediction head.

Table 4.1: Varied factors and matched controls.

Category	**Treatment**
Aggregation method	Primary varied factor (three levels)
Label condition and corruption level	Robustness factors
Catalog-label pattern	Subgroup analysis
Molecular encoder	One pretrained, frozen artifact shared by all methods
Prediction head	Trained separately for each method; input size differs (concatenation uses 512 inputs)
Partitions and inputs	Same across methods
Loss, optimizer, stopping rule, training budget	Same across methods
Random seeds and corruption draws	Same seed IDs across methods; variation across seeds is reported separately (Section 4.8)

4.2 Data Acquisition and Preparation

This section describes the data sources, how they were joined, and how the model inputs (molecular structures) and targets (scent labels) were prepared. Molecular structures were converted into graphs, and scent labels were organized into a fixed set of labels.

4.2.1 Data Sources and Provenance

The study used four kinds of records (Table 4.2). The single-molecule sources are GoodScents and Leffingwell, both accessed through Pyrfume. They are independently maintained reference catalogs for the perfumery and flavor industries, and each record lists a molecule, its structure, and the scent labels assigned to it. The Leffingwell labels are described as assigned by olfactory experts, usually practicing perfumers [23]. The blend data came from the public repository of Sisson et al. [37], and archived GoodScents pages supplied the direction of each blending recommendation.

Structure filtering. Before molecules were matched across sources, structures were sanitized with RDKit: salts were removed, molecules containing elements other than C, H, O, N, S, F, Cl, Br, and I were excluded, and molecules outside 20–600 g/mol were excluded. These rules are inclusion heuristics for typical aroma chemicals. This step used structure only; no label was consulted.

Join key. The two single-molecule catalogs were merged into one master dictionary using the isomeric SMILES string as the key. Every SMILES string in both catalogs and in the blend dataset was canonicalized with RDKit so that the same molecule written in two ways matched correctly. SMILES was chosen over the PubChem CID because some CIDs in the source files are negative or

Table 4.2: Data sources.

Source	**Data organization**	**Purpose in this study**
GoodScents (single molecules)	Scent labels per stimulus; a stimulus-to-CID metadata table; a CID-to-structure table	Reference labels and pretraining; source-consistency check
Leffingwell (single molecules)	One indicator column per scent label for each stimulus, joined through stimulus and CID	Reference labels and pretraining; source-consistency check
Sisson export (blend pairs)	Two component columns (CID.1 and CID.2) with a list of scent labels for each pair; component identities were matched and duplicate pairs removed	Blend targets
Archived GoodScents pages	Each component page lists suggested blending partners under scent label headings; recommendations were collected while recording which component made each suggestion	Evidence for two-page scent label intersection (Section 4.3)

non-standard, and SMILES must be canonicalized anyway when molecules are converted to graphs. Blend components in the Sisson et al. dataset, which are identified by CID, were linked to the master dictionary by canonicalizing their SMILES in the same way. Duplicate records, components that could not be joined or were filtered out, self-pairs, and scent labels with no match in the vocabulary were handled at this step; their counts are reported in Chapter 5.

Blend records. The Sisson et al. blend data came from blender recommendations crawled from GoodScents molecule pages [37]. Each page lists recommended partners under scent label headings, and because the two molecules' pages were crawled separately, the same pair and scent label can appear from one direction or from both. This study kept the direction of each listing so that the two-page scent label intersection (Section 4.3) could be computed.

4.2.2 Catalog Targets and Their Limits

Catalog-recorded scent labels serve as proxy targets for supervised learning. They are curated reference records with some noise and subjectivity [6]. This study treats them as a practical proxy for perceived scent, estimates how consistent they are, and examines how model comparisons change as label reliability changes.

4.2.3 Target Vocabulary

Two vocabularies are kept apart. The *standardized reference vocabulary* has 104 scent labels (Appendix A.2.3), obtained after spelling variants were merged, and is used for pretraining. The *final target vocabulary*, of size L, is the subset of these scent labels on which the blend models are trained and scored.

Standardization. The raw GoodScents and Leffingwell labels contain spelling variants of the same scent label (for example, anisic and anise, or caramel and caramellic), so this study adopted the open-source canonicalization dictionary of Sisson et al. [37]. scent labels the dictionary did not cover were inventoried and reviewed.

Final target vocabulary. The final vocabulary was built from training data only (Section 4.5.4). Validation and test labels were not consulted when deciding which scent labels were kept.

Molecules without reference labels. Training molecules with no mapped reference labels kept their structures, and their targets were marked unknown (masked). The model has no "unknown" or "odorless" class.

4.3 Label Reliability Mask

Not every recorded scent label is equally trustworthy, and not every missing label is a true absence. For each blend–scent label cell (one blend and one scent label in the label matrix), the study decided whether that cell is used in the stricter eligibility-mask condition.

4.3.1 Consistency Evidence

Two kinds of evidence are used to measure consistency.

- **Two-page scent label intersection (cell level).** For a pair of molecules A and B, let E_A be the scent labels under which A's page recommends B, and let E_B be the scent labels under which B's page recommends A. A scent label has primary support for the pair when it is in both, $E_A \cap E_B$. For example (illustrative), if $E_A = \{\text{sweet, fruity}\}$ and $E_B = \{\text{fruity, woody}\}$, only "fruity" is supported. Requiring both pages reduces reliance on one-sided recommendations,

at the cost of excluding valid labels when the reverse recommendation is missing or a page is incomplete, and both pages come from the same catalog.

- **scent label-level consistency (scent label level).** Agreement between GoodScents and Leffingwell was measured for each scent label using Cohen's kappa, computed only on training-partition molecules that occur in both catalogs. A scent label with $\kappa \geq 0.21$ is called consistent.

4.3.2 Eligibility Mask

Table 4.3 defines the eligibility mask $M_{ij} \in \{0, 1\}$ for blend i and scent label j.

Table 4.3: Cell-level rules for the eligibility mask.

Target	**scent label κ**	**Page support**	**Interpretation**	**Mask**
Listed	≥ 0.21	2 (both pages)	Reliable positive	M = 1
Listed	≥ 0.21	1 (one page)	Not confirmed	M = 0
Listed	≥ 0.21	0 (not recovered)	Support unknown	M = 0
Listed	< 0.21	Any	Agreement too weak	M = 0
Unlisted	≥ 0.21	Not applicable	Eligible negative (closed-world assumption)	M = 1
Unlisted	< 0.21	Not applicable	Negative status not reliable	M = 0

Because eligible unlisted scent labels enter as negatives, the masked test set contains both positives and negatives, so a model that predicts an unrecorded scent label is penalized as a false positive.

Closed-world assumption. Unlisted scent labels are treated as absent so that they can serve as negative labels for training and evaluation. A catalog may simply have left a scent label out, which the eligibility mask reduces but does not remove.

4.3.3 Experimental Label Conditions

Three label conditions are used:

- **Primary (eligibility-mask) condition.** Only cells with $M_{ij} = 1$ under Table 4.3 are used.
- **Expanded either-page sensitivity.** Same as the primary condition, except that listed positives with a page support of 1 are also counted as positives, to test how much the results depend on the stricter primary requirement.

- **Full-label diagnostic.** Every cell is used ($M_{ij} = 1$ for all i, j), with listed scent labels as positives and unlisted scent labels as negatives.

Masked loss. Masking excludes cells from the loss and scoring; in the primary condition a blend is kept as long as it has at least one eligible cell. The loss is computed only over eligible cells:

$$L_{\text{masked}} = \frac{\sum_i \sum_j M_{ij}\, \ell_{ij}}{\sum_i \sum_j M_{ij}} \tag{4.1}$$

where ℓ_{ij} is the per-cell loss (Section 4.4.3). With $M_{ij} = 1$ everywhere this is the ordinary mean loss. The same mask is applied when metrics are computed (Section 4.7).

Limited support. A scent label is included in macro-averaged scores only if the evaluated partition has at least one eligible positive and at least one eligible negative cell for it. This working rule was adopted before the main model results were evaluated, in place of the five-per-side rule of the approved manuscript. Excluded scent labels are listed in the report, and the output layer keeps all L outputs.

4.4 Model Architecture and Training

All three models use the same frozen molecular graph encoder, and each has its own separately trained prediction head. They differ only in the step that combines the two molecule embeddings. Each molecule is turned into a graph and encoded into a vector, the two vectors are combined by the aggregation method, and the combined vector is converted into one probability per scent label.

4.4.1 Shared Graph Encoder

Each molecule is a graph in which atoms are nodes and bonds are edges. A Graph Attention Network (GAT) encoder turns each graph into a 256-dimensional vector $\mathbf{z}$, so a blend yields two embeddings $\mathbf{z}_A$ and $\mathbf{z}_B$, and the same encoder weights are applied to both molecules.

The encoder receives the 46-dimensional atom features and 10-dimensional bond features listed in Appendix A and uses three edge-aware GATv2 layers. The first two layers use four attention heads with 128 features per head, and the final layer uses one head to produce a 256-dimensional node representation. A residual connection is used in the second layer, followed by an attention readout that gives one vector per molecule. The attention computation is defined in Section 3.1.3.

Pretraining. The encoder was first pretrained on single-molecule data from the training partition only (Section 4.5), with a portion set aside for internal validation. A temporary head with 104 sigmoid outputs (one per standardized reference scent label) was trained with masked binary cross-entropy, so molecules with unknown targets did not contribute to the loss. The best checkpoint was restored, the temporary head was removed, and the encoder weights were saved and frozen.

Frozen and trainable parts. The encoder is frozen during the blend-level comparison, so all three variants receive embeddings from the same encoder weights and only the aggregation module and prediction head are trained. This keeps the comparison focused on how the two embeddings are combined.

4.4.2 Aggregation Variants

Table 4.4 summarizes the three variants.

Table 4.4: The three aggregation variants.

Variant	**Operation**	**Order handling**	**Head input**
Mean pooling	Average the two molecule embeddings	Invariant by construction	256
Order-averaged concatenation	Run both ordered concatenations through the head, average the logits, then apply one sigmoid	Explicit symmetrization	512 (two passes)
Set Transformer aggregation	Two self-attention blocks (SAB), then pooling by multihead attention (PMA)	SAB is equivariant; PMA gives an invariant output	256

Mean pooling. The two embeddings are averaged to produce one blend embedding:

$$\mathbf{z}_{\text{blend}} = \tfrac{1}{2}(\mathbf{z}_A + \mathbf{z}_B) \in \mathbb{R}^d. \tag{4.2}$$

Order-averaged concatenation. The two embeddings are placed side by side, $[\mathbf{z}_A \,|\, \mathbf{z}_B] \in \mathbb{R}^{2d}$, and the model is run on both (A, B) and (B, A). The two logits are averaged before one sigmoid is applied. This adapts the pairwise design of Sisson et al. [37] and uses this study's encoder, loss, and partitions.

Set Transformer aggregation. The pair $\{\mathbf{z}_A, \mathbf{z}_B\}$ is treated as a set and passed through two self-attention blocks (SABs), each with four attention heads and a 512-unit feedforward layer.

A pooling by multihead attention (PMA) layer with one learned seed vector then combines the two outputs into a single blend embedding. The implementation is based on the Set Transformer modules of Lee et al. [24].

The number of trainable parameters and the training and inference time of each model are reported with the results.

4.4.3 Output Head and Loss

Each prediction head is a multilayer perceptron mapping $256 \to 128 \to 92$, where $L = 92$ is the number of target scent labels (Section 4.2.3). For order-averaged concatenation the first layer takes 512 inputs. The output layer applies an independent sigmoid to each of the L outputs, so scent labels are predicted independently. Blend-level training uses the asymmetric focal loss of Equation 3.9 with $\gamma_{\text{pos}} = 0$, $\gamma_{\text{neg}} = 2$, a negative margin of 0.05, no class-frequency weights, and a mean over eligible cells; its per-cell values enter the masked loss of Section 4.3.3. Binary cross-entropy is used only for encoder pretraining.

Table 4.5: Blend-level training settings (same for all three variants).

Setting	**Value**
Optimizer	Adam
Learning rate	0.001
Weight decay	0
Learning-rate schedule	Constant
Batch size	128
Gradient clipping	1
Early stopping	Patience of 20 epochs; minimum improvement of 0.0001
Maximum epochs	200
Loss	Asymmetric focal ($\gamma_{\text{pos}} = 0$, $\gamma_{\text{neg}} = 2$, margin 0.05)

Only the focal exponents and the five seeds were specified in the approved manuscript; the remaining settings are fixed development defaults and were not tuned.

Seeds and checkpoints. Each variant was trained with the same five seeds (0 to 4), which change the prediction head's initialization and the order of the training data. For each run, the checkpoint with the lowest validation loss was saved and used for testing. Early stopping decides when training ends, and the lowest validation loss decides which epoch's weights are kept.

4.5 Component-Disjoint Partitioning and Graph Carving

This section explains how the data were divided into training, validation, and test sets so that no molecule in a training blend also appears in a validation or test blend.

4.5.1 Scaffold Check

Scaffold splitting groups molecules by shared core framework. To check whether it would be informative on the present data, the molecule-to-scaffold ratio was computed:

$$R = \frac{N_{\text{molecules}}}{N_{\text{scaffolds}}} \tag{4.3}$$

where $N_{\text{molecules}}$ is the number of unique canonical molecules and $N_{\text{scaffolds}}$ is the number of distinct Bemis–Murcko frameworks produced by RDKit. A value of R near 1 means almost every scaffold holds a single molecule. R was computed on the merged master dictionary and compared with the threshold $R < 2.0$ reported by Fooladi et al. [12]. The resulting ratio is reported in Chapter 5.

4.5.2 Cluster-Based Partitions

Molecules were grouped by structural similarity, and whole groups were assigned to one partition. Each unique molecule was represented by a 2048-bit binary ECFP4 fingerprint with radius 2. UMAP projected the fingerprints to two dimensions using Jaccard distance, 15 neighbors, a minimum distance of 0.1, and random seed 42. K-Means then formed $k = 7$ clusters using 20 initializations and random seed 42, following Guo et al. [15]. UMAP's default approximate neighbor search left two very small structures disconnected, so exact Jaccard nearest neighbors were computed and supplied to UMAP, and disconnected components were retained (infinite disconnection distance).

Clusters were assigned to training, validation, and test partitions using molecule counts only. All assignments giving at least one cluster to each partition were enumerated, and the one with the smallest total absolute deviation from a 70:15:15 molecule ratio was selected, with ties broken lexicographically by cluster identifier. The rule used no scent label identities, label frequencies, pair counts, or test labels. Once fixed, the partitions were not changed. The cluster-to-partition assignment, molecule counts, retained pairs, and scent label support are reported in Chapter 5.

4.5.3 The Graph-Carving Protocol

Each molecule was assigned to a partition based on its cluster. A blend was retained only if both of its molecules belonged to the same partition, and blends connecting molecules from different partitions were discarded. As a result, no molecule appears in more than one partition. Structurally similar molecules can still fall in different partitions if they belong to different clusters, and a molecule can appear in multiple blends within its own partition.

The validation set was used for early stopping, hyperparameter decisions, and selection of the decision threshold. The test set was kept separate and used only for the final evaluation.

4.5.4 Training-Only Vocabulary Filter

Carving changes how often each scent label occurs, so the final target vocabulary was chosen after partitioning, from training data only. A standardized scent label d was kept if it had at least $x = 30$ positive training blends:

$$d \in D_{\text{target}} \iff c_{\text{train}}(d) \geq x \tag{4.4}$$

where $c_{\text{train}}(d)$ is the number of training blends in which d is recorded as positive. The vocabulary was determined once from the full-label training partition and is shared across all label conditions, so changing the eligibility mask does not change the prediction targets. This gives the final vocabulary size L. The numbers of positive and negative labels per scent label in the validation and test sets are reported separately and were not used to select the vocabulary. scent labels with insufficient eligible test support are handled by the rule in Section 4.3.3.

4.6 Experiments

The three aggregation methods use the same partitions, encoder, loss function, optimizer, stopping rule, training budget, and evaluation metrics, and each is trained with the same five seeds (0, 1, 2, 3, and 4).

The main experiments consist of 3 aggregation methods $\times$ 5 seeds $\times$ 3 label conditions, giving 45 model runs. The controlled corruption experiment adds another 45 runs and reuses the 15 clean primary-condition runs as its reference.

4.6.1 Experiments A and B: Main Comparison (RQ1)

Experiment A trains each model in the full-label condition, and Experiment B trains each model in the primary eligibility-mask condition (Section 4.3.3). A third run uses the expanded either-page condition as a sensitivity check. All are evaluated on the held-out test partition and scored on the cells defined by their own mask, which shows how much the comparison depends on the label condition.

4.6.2 Experiment C: Controlled Label Corruption (RQ3)

To examine how the comparison changes as training labels become less reliable, labels were corrupted on purpose in the training partition only; validation and test labels were never changed. Corruption was applied to the eligible cells of the primary condition. For a noise level $p \in \{0.10, 0.20, 0.30\}$ and N eligible positive cells, $\lfloor pN \rfloor$ positives were flipped to absent and the same number of eligible negatives were flipped to present, which keeps the number of positives unchanged. Flips were drawn across the whole training matrix, not per scent label. For each noise level and seed, one corruption pattern was generated from a random stream seeded by the level and the seed, and shared across all three aggregation methods. The $p = 0$ reference is the uncorrupted Experiment B run. Models were retrained at each level and scored on the unchanged validation and test labels. The change in macro F1 from $p = 0$ to $p = 0.30$ is compared across methods with the paired bootstrap of Section 4.8.

4.6.3 Blend-Type Analysis (RQ2)

Each test blend is compared with the recorded scent labels of its two parent molecules, taken as the union of the two components' standardized GoodScents and Leffingwell scent labels, and assigned to one of three groups:

- **Catalog-label loss:** a scent label recorded for either parent is missing from the blend.
- **Catalog-label gain:** the blend has a recorded scent label that neither parent has.
- **Additive:** the blend shows neither loss nor gain.

The number of blends in each group is reported, and model performance is compared separately within each group. This is an exploratory subgroup analysis. If a group contains fewer than 5%

or more than 95% of the test blends, blends are instead grouped by the Jaccard overlap $J = |P \cap B|/|P \cup B|$ between the parents' scent label union P and the blend's scent labels B. Three fixed equal-width bins are used: low ($0 \leq J < 1/3$), middle ($1/3 \leq J < 2/3$), and high ($2/3 \leq J \leq 1$), with boundary values assigned to the upper bin. The bins were fixed before subgroup scoring and are not adjusted when a bin is sparse. Macro metrics are reported as unavailable for a bin without eligible positives.

4.7 Evaluation Metrics

4.7.1 Multi-Label Metrics

Let TP be a scent label correctly predicted present, FP a scent label incorrectly predicted present, and FN a present scent label that was missed.

- **Macro F1 (primary metric).** F_1 is computed for each scent label separately and then averaged with equal weight, so rare scent labels count as much as common ones [30].
- **Micro F1.** TP, FP, and FN are pooled across all scent labels before F_1 is computed; it favors common scent labels.
- **Average precision (PR-AUC).** The step-wise average precision (as in scikit-learn's average_precision_score, not trapezoidal integration of the precision–recall curve), which summarizes performance across all probability thresholds. Macro AP averages the per-scent label values over the scored scent labels, and micro AP is computed over all eligible cells pooled.
- **Hamming loss.** The fraction of evaluated cells that are predicted incorrectly.

Decision cutoff. For each method and label condition, one global cutoff converts predicted probabilities into present or absent labels. It is selected from the grid $\{0.05, 0.10, \ldots, 0.95\}$ as the value giving the highest validation macro F1 averaged across the five seeds, and is then applied unchanged to all five seeds at test.

4.7.2 Masked Metrics

All metrics are computed only over cells with $M_{ij} = 1$, using the mask of the condition being evaluated (all ones in the full-label condition). Macro metrics use only scent labels with enough

eligible support in the evaluated partition (Section 4.3.3). With a mask, Hamming loss is

$$\mathrm{HL}_M = \frac{\sum_{i=1}^{N} \sum_{j=1}^{L} M_{ij}\, \mathbf{1}(y_{ij} \neq \hat{y}_{ij})}{\sum_{i=1}^{N} \sum_{j=1}^{L} M_{ij}} \tag{4.5}$$

where N is the number of evaluated blends, L is the number of target scent labels, y_{ij} is the label, and $\hat{y}_{ij}$ is the prediction after applying the decision cutoff. Smaller values are better.

All-negative reference. Hamming loss rewards the many easy negatives, so each model's value is reported next to that of a baseline predicting every evaluated cell as absent. This baseline equals the fraction of evaluated cells that are positive:

$$D_M = \frac{\sum_i \sum_j M_{ij}\, y_{ij}}{\sum_i \sum_j M_{ij}} \tag{4.6}$$

D_M is computed separately for each partition and mask. It differs from mask coverage, the fraction of cells with $M_{ij} = 1$.

4.8 Statistical Comparison

Differences between models can reflect chance in which blends happen to be in the test set, or chance in training. The two sources are handled separately.

Variability across seeds. Each model was trained with the same five seeds (0 to 4). Seed-to-seed variation is summarized separately as the mean and spread across runs. Within each bootstrap replicate, every seed is evaluated separately on the resampled test blends, and the paired differences between two methods across the five matching seeds are then averaged.

Bootstrap over blends. scent label outcomes within one blend are related, so individual cells are not sampled independently. Whole test blends are resampled as rows, each keeping all its scent label labels and eligibility mask. From the N test blends, N rows were drawn with replacement 10,000 times, using random seed 42. All models and seeds are scored on the same resampled blends, giving a paired comparison. For replicate b, the difference between two methods is

$$\Delta(b) = \frac{1}{5} \sum_{s=0}^{4} \left[\mathrm{Metric}_{\text{model 1},\, s}(b) - \mathrm{Metric}_{\text{model 2},\, s}(b) \right], \tag{4.7}$$

where s indexes the seeds. Higher values are better for macro F1, micro F1, and PR-AUC, and lower values are better for Hamming loss. The comparison is made for each pair of aggregation

methods (three pairs), and confirmatory inference is limited to the three macro-F1 comparisons. For each pair, the two-sided bootstrap probability is twice the smaller of the proportions of $\Delta(b)$ values at or below zero and at or above zero, with a plus-one correction, capped at one. The three probabilities are adjusted with the Holm procedure at a family-wise significance level of $\alpha = 0.05$. The other metrics are reported with effect sizes and confidence intervals only.

Scored labels and zero denominators. The scent labels included in macro averages are determined once on the original test split and stay the same across all bootstrap replicates. If a resample contains no positive cells for an included scent label, its F1 and average precision are set to 0.0; the scent label stays in the macro average and the resample is not redrawn. Micro F1, micro AP, and Hamming loss are computed over all eligible cells pooled.

Intervals and effect size. A 95% percentile interval is taken from the 2.5th and 97.5th percentiles of the 10,000 differences. The effect size is the mean paired difference, in the units of the metric. Test blends that share a molecule are related, so these intervals may underestimate the true uncertainty.

Appendix A

Chemical Feature Vectors

The exact composition of the 46-dimensional node feature vector and the 10-dimensional edge feature vector used for the intra-molecular GAT encoder are detailed below.

Chemical Property	**Dimensions**	**Mathematical / One-Hot Representation & Details**
Atomic Identity	9	One-hot encoding of key organic and halogen elements in flavor/fragrance compounds: [C, N, O, S, F, Cl, Br, I, Other].
Formal Charge	1	Scalar representation of the localized net electrical charge on the atom.
Atomic Degree	6	One-hot representation of the number of connected covalent bonds to heavy atoms: [0, 1, 2, 3, 4, 5+].
Explicit Valence	7	One-hot encoding of the total valence electrons committed to bonding in the molecule: [0, 1, 2, 3, 4, 5, 6+].
Hybridization State	6	One-hot representation of the orbital configuration: [sp, sp^2, sp^3, sp^3d, sp^3d^2, Other].
Aromaticity	1	Binary flag indicating if the atom is part of a resonance-stabilized aromatic ring structure.
Attached Hydrogens	5	One-hot encoding of the number of hydrogens bound to this heavy atom: [0, 1, 2, 3, 4+].
Ring Size Participation	6	Binary flags representing specific ring sizes to map cyclic functional groups: [3-ring, 4-ring, 5-ring, 6-ring, 7-ring, 8-ring].
Ring Membership	1	Binary flag indicating whether the atom is enclosed within any ring structure.
Chirality Tag	4	One-hot representation of tetrahedral stereocenter configuration: [Unspecified, Clockwise (CW), Counterclockwise (CCW), Other].

Table A.1: Specification of the 46-Dimensional Atom (Node) Feature Vector (X_i).

Chemical Property	**Dimensions**	**Mathematical / One-Hot Representation & Details**
Bond Type	4	One-hot representation of the bond's electronic sharing structure: [Single, Double, Triple, Aromatic].
Conjugation	1	Binary flag indicating if the bond belongs to a continuous, overlapping pi-electron system.
Ring Membership	1	Binary flag indicating if the bond is closed within a ring structure.
Stereochemical Config	4	One-hot representation of spatial configuration around double bonds: [None, Stereo-Z, Stereo-E, Other].

Table A.2: Specification of the 10-Dimensional Bond (Edge) Feature Vector (E_{ij}).

A.1 Symmetry Collapse: Representation-Theoretic Derivation

This appendix provides the fuller representation-theoretic argument for a possible expressivity limitation of linear, order-independent readouts at the mixture level, summarised in the mean-pooling discussion of the main text.

Consider a molecular graph G whose set of node embeddings M occupies a d-dimensional representation space $\mathcal{U}$. The structural symmetries of the graph are captured by its automorphism group $\mathcal{H}$, which collects all permutations of the node indices that leave the graph structure unchanged. When a linear, $\mathcal{H}$-invariant readout function f is applied to aggregate these embeddings, representation theory requires that it factor through the group-averaging Reynolds operator p_{avg}, defined as [41]:

$$p_{\text{avg}} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \rho(h). \tag{A.1}$$

The Reynolds operator projects the full representation space $\mathcal{U}$ onto its trivial, symmetry-fixed subspace $\mathcal{U}_1$. As a consequence, any linear $\mathcal{H}$-invariant map decomposes into a downstream linear map composed with this group-averaging projection [41]:

$$f = f' \circ p_{\text{avg}}. \tag{A.2}$$

This factorization establishes that a linear readout can only operate on the information that survives in $\mathcal{U}_1$. The non-trivial, symmetry-sensitive components $\mathcal{U}_2 \oplus \cdots \oplus \mathcal{U}_m$, which encode the node-to-node variation responsible for structural differences between molecules, can be discarded by the projection [41]. Thus, two distinct combinations that share the same element-wise sum or mean can become indistinguishable to a downstream classifier. This is a possible information loss in the representation; it does not show that such collisions dominate the present dataset. The result therefore motivates testing attention-based aggregation alongside mean pooling, not replacing mean pooling or assuming that attention will perform better [37, 41].

A.2 Dataset Profiles and Canonical Vocabulary

A.2.1 Single-Molecule Harmonized Dataset Sample

Table A.3 provides representative single-molecule entries resulting from the consolidation of the GoodScents and Leffingwell databases under RDKit organic sanitization and canonicalization.

Table A.3: Representative sample of harmonized single odorants.

CID	Common Name	Canonical SMILES	Harmonized Descriptors
1183	Vanillin	`COc1cc(C=O)ccc1O`	chocolate, creamy, sweet, vanilla
1254	Menthol	`CC1CCC(C(C)C)C(O)C1`	cooling, mentholic, minty, woody
3314	Eugenol	`C=CCc1ccc(O)c(OC)c1`	cinnamon, clove, phenolic, spicy, woody
10582	Myrtenol	`CC1(C)C2CC=C(CO)C1C2`	balsamic, camphoreous, herbal, pine
31276	Isoamyl acetate	`CC(=O)OCCC(C)C`	banana, estery, fruity, pear, sweet
87630	2-Decen-1-ol	`CCCCCCCC=CCO`	citrus, fatty, floral, rose, waxy
637511	Cinnamaldehyde	`O=C/C=C/c1ccccc1`	candy, cinnamon, spicy, warm

A.2.2 Pairwise Blend Mixture Sample

Table A.4 presents sample binary mixtures from the pairwise blend benchmark dataset compiled by Sisson et al. [37], illustrating the constituent molecules and their catalog-recorded blend descriptors.

Table A.4: Sample pairwise odor mixtures from the Sisson et al. benchmark with catalog-recorded descriptors.

Blend	Component 1 (CID)	Component 2 (CID)	Catalog-Recorded Descriptors
B-01	Acetaldehyde (177)	2-Heptanol (10976)	citrus, fruity, pungent
B-02	Acetoin (179)	Bornylcyclohexanol (111394)	buttery, creamy, woody
B-03	2-Oxobutanoic acid (58)	Methyl dihydrojasmonate (102861)	brown, caramellic, floral
B-04	2-Oxobutanoic acid (58)	γ-Undecalactone (7714)	brown, caramellic, fruity
B-05	Pentopyranose (229)	Perillaldehyde (16441)	herbal, spicy
B-06	2-Oxobutanoic acid (58)	Butyric acid (264)	cheesy, sour
B-07	Cinnamaldehyde (637511)	Eugenol (3314)	spicy

A.2.3 Standardized 104-Odor Descriptor Vocabulary Gate

Table A.5 enumerates the complete 104-descriptor standardized vocabulary gate enforced during Phase 1 preprocessing.

Table A.5: The 104 standardized odor descriptors (alphabetical).

1. acetic	27. clean	53. honey	79. potato
2. acidic	28. cocoa	54. jammy	80. powdery
3. alcoholic	29. coconut	55. juicy	81. pungent
4. aldehydic	30. coffee	56. lactonic	82. roasted
5. alliaceous	31. cooling	57. leathery	83. rooty
6. amber	32. corn	58. licorice	84. rummy
7. ammoniacal	33. coumarinic	59. malty	85. salty
8. animal	34. creamy	60. marine	86. smoky
9. anise	35. dairy	61. meaty	87. soapy
10. aromatic	36. dusty	62. medicinal	88. solvent
11. balsamic	37. earthy	63. melon	89. sour
12. berry	38. eggy	64. mentholic	90. spicy
13. bitter	39. estery	65. minty	91. sulfurous
14. bready	40. ethereal	66. moldy	92. sweet
15. brown	41. fatty	67. mossy	93. tarragon
16. burnt	42. fermented	68. mushroom	94. thujonic
17. buttery	43. fishy	69. musk	95. toasted
18. cabbage	44. floral	70. mustard	96. tobacco
19. camphoreous	45. fresh	71. musty	97. tomato
20. caramellic	46. fruity	72. nutty	98. tonka
21. celery	47. fungal	73. oily	99. tropical
22. cheesy	48. fusel	74. onion	100. vanilla
23. chemical	49. garlic	75. orris	101. vegetable
24. cherry	50. green	76. peach	102. waxy
25. chocolate	51. hay	77. phenolic	103. winey
26. citrus	52. herbal	78. pine	104. woody

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